Roger Truong

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EDUCATION

University of New South Wales — **Sydney, New South Wales**
B.S. in Computer Science — *Jan 2023 - May 2026*

- **Grade:** Distinction

EXPERIENCE

AndCo (YC P26): AI Personal Injury Legal Platform (Python | FastAPI | React | PostgreSQL) — **San Francisco, CA**
Software Engineer *(FDE @ Lyra)* — *Sep 2026 - Current*

- Enabled **automated portal authentication across 4 verification methods** - email, SMS, TOTP and authenticator-app approvals, by building a **FastAPI MFA service** that coordinates browser agents with Android devices, Redis-backed email code delivery and secure credential management.
- Prevented premature follow-ups across **4 AI request agents** handling medical records, police reports, insurance claims and coverage, by implementing **event-driven coordination**, pending-document review gates and a shared, validated tool-call contract.
- Established an **Australian staging environment spanning 3 web applications, an API and a background worker** by building **GitHub Actions/GCP Cloud Run** deployment workflows with isolated Neon databases, migration rehearsals and automated readiness checks.

Corveris: AI Crime Reporting Platform (Python | FastAPI | React | Next.js | Terraform) — **San Francisco, CA**
Software Engineer *(FDE @ Lyra)* — *Jan 2026 - Aug 2026*

- Enabled **one-click verification of AI-generated report summaries** by implementing **typed citation schemas**, transcript turn-ID mapping and **Unicode-aware text-span matching**, with memoised reverse indexes supporting **bidirectional navigation and precise source highlighting** across English and original-language transcripts in the React dashboard.
- Improved **report accuracy by 25%** and reduced **officer review time by 30 minutes per incident**, by building **jurisdiction-specific penal code resolution** for California and Texas through **YAML-based configuration**, **NIBRS-compliant classification overrides**, and **LLM-powered sub-classifiers** for theft, fraud, vandalism, assault and hit-and-run incidents.
- Increased **dashboard load performance by 45%** for tables with **400+ active reports**, by decoupling **paginated report queries from full-table metadata scans**, introducing **PostgreSQL indexes and server-side aggregation caching**, and implementing **TanStack Query** caching with retained page data alongside **TanStack Virtual** table rendering.
- Reduced **anonymous report fraud by 70%** while maintaining a seamless citizen experience, by implementing a **claim-via-SMS workflow** using **Twilio OTP verification**, **rate-limited, single-use claim tokens with 72-hour expiry**, and **constant-time comparison** for URL-safe Base64 identity binding.

Cognimark: Dementia Assistance Platform (Python | AWS SAM | Django | Vue.js) — **Boston, MA**
Software Engineer *(FDE @ Lyra)* — *Oct 2025 – Dec 2025*

- Reduced **EHR data duplication by 40%** across **3 healthcare organisations**, saving an estimated **$15,000/year** in wasted storage and manual reconciliation, by designing **deterministic FHIR R4 deduplication logic** within **AWS Step Functions** ingestion pipelines that cross-referenced patient records before persisting to per-tenant **PostgreSQL** databases.
- Accelerated **patient data ingestion throughput by 60%** for multi-tenant healthcare clients, by architecting a **serverless pipeline (API Gateway, Step Functions, Lambda, S3)** with **resumable job tracking**, fault-tolerant NDJSON assembly, and batch conflict resolution via **HTTP 409 retry logic**.
- Delivered a **full-stack clinical management application** integrated by **3 organisations** such as AthenaHealth to manage **5,000+ patient records**, by building a **Django REST Framework** backend with **Celery** async task queues and a **Vue 3** frontend using TanStack Query and Pinia, integrated with **AWS Cognito JWT authentication** and **multi-tenant database routing**.

ACTIVITIES AND LEADERSHIP

UNSW Teochew Student Association — **Sydney, New South Wales**
President — *Dec 2023 – Current*

- Leader of an organisation of **500+ members**, **organising** on-campus and off-campus **social events**, **managing** a board of executives, spreading awareness of Chinese Dialect nation-wide through **social media** campaigning.
- Acting as a **role model and taking responsibility for the delegation** and completion of tasks on time.

SKILLS

Programming: Java, Rust, C, C++, Python, JavaScript/TypeScript, HTML/CSS, SQL, POSIX Scripting, C#
Tools: Git, React.js, Next.js, Flask, Node.js, Express.js, Spring Boot, Maven, Gradle, Angular